Mine Economics

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Economic importance of the mineral industry

Mining economy,

Risky nature of the mining industry,

Demand and Supply,

Elasticity of Demand,

National mineral policy.

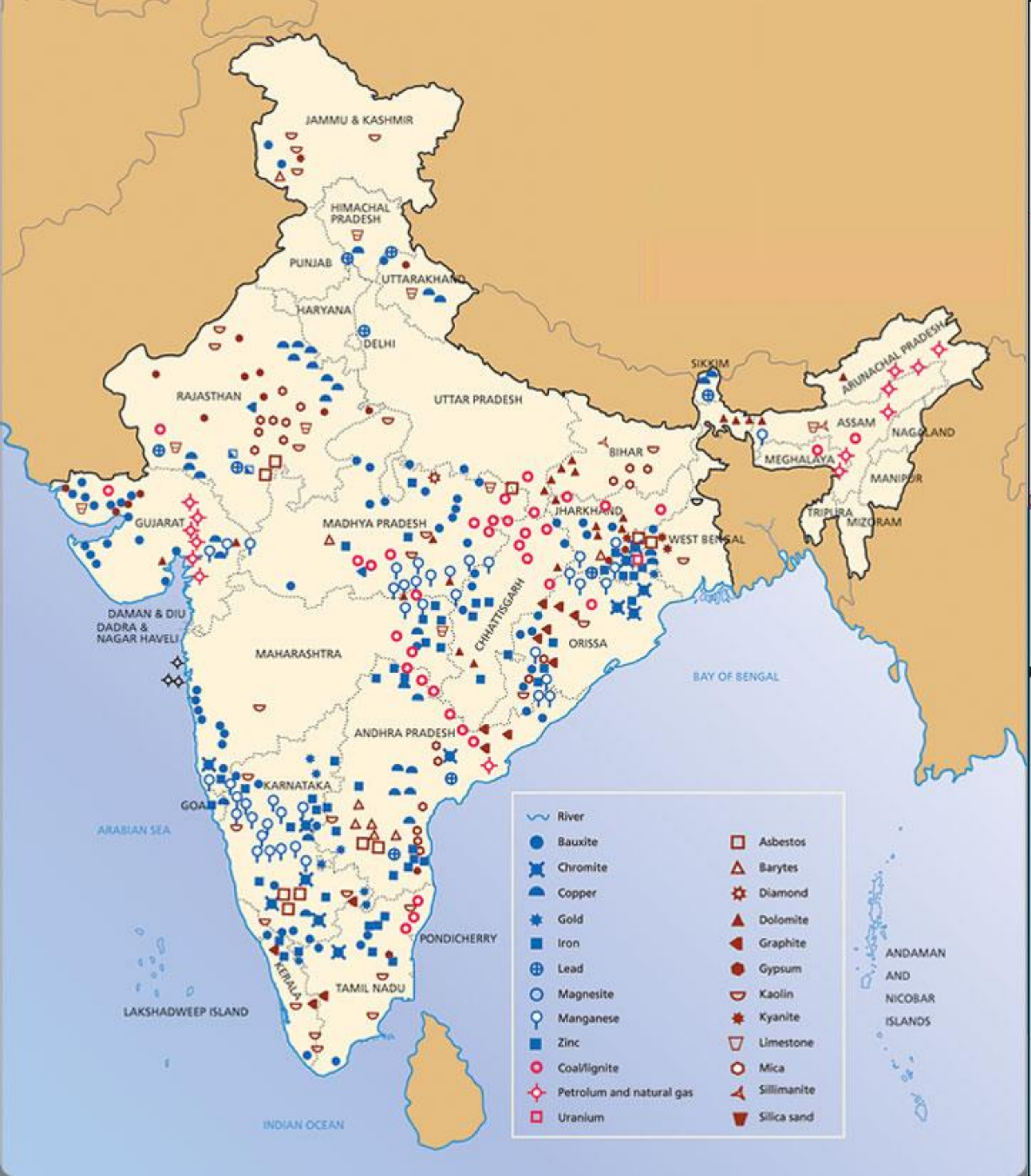
- In a general sense, **economics** is the study of production, distribution, and consumption and can be divided into two broad areas of study: macroeconomics and microeconomics. **Macroeconomics** deals with aggregate economic quantities, such as national output and national income. Macroeconomics has its roots in **microeconomics**, which deals with markets and decision making of individual economic units, including consumers and businesses. Microeconomics is a logical starting point for the study of economics.

Introduction

India produces as many as 87 **minerals**, which includes 4 fuel, 10 metallic, 47 non-metallic, 3 atomic and 23 minor **minerals** (including building and other materials) and is the world's largest producer of mica blocks and mica splitting.

In 2019, the country was the 4th largest world producer of iron ore; 4th largest worldwide producer of chromium; 5th largest world producer of bauxite; 5th largest world producer of zinc;7th largest producer of manganese in the world; 7th largest producer of lead in the world; 7th largest producer of sulfur in the world; 11th largest world producer of titanium; 18th largest world producer of phosphate; 16th largest world producer of gypsum; 5th largest world producer of graphite; 3rd largest world producer of salt. It was the 11th the world's largest producer of uranium in 2018

- A resource is something that is useful and valuable in the condition in which we find it. In its raw or unmodified state it may be an input into the process of producing something of value, or it may enter consumption directly and thus be valued as an amenity
- the major classes of natural resources:
- "...(1) Agricultural land; (2) forest land and its multiple products and services; (3) natural land areas preserved for esthetic, recreational, or scientific purposes; (4) the fresh and salt water fisheries; (5) mineral resources that include the mineral fuels and nonfuel; (6) the renewable non-mineral energy sources of solar, tidal, wind, and geothermal systems; (7) water resources; and (8) the waste-assimilative capacities of all parts of the environment...“
- mineral resources, as commonly used by mineral industry people, is "...A concentration of naturally occurring solid, liquid, or gaseous materials in or on the earth's crust in such form that economic extraction of a commodity is currently or potentially feasible...“
- Reserves (sometimes also called ore) include minerals in deposits fulfilling all three conditions -- they are geologically known (quantity and grade), their production is economically feasible at the present, and the technology for their extraction, ore processing, and use is currently available.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Manganese
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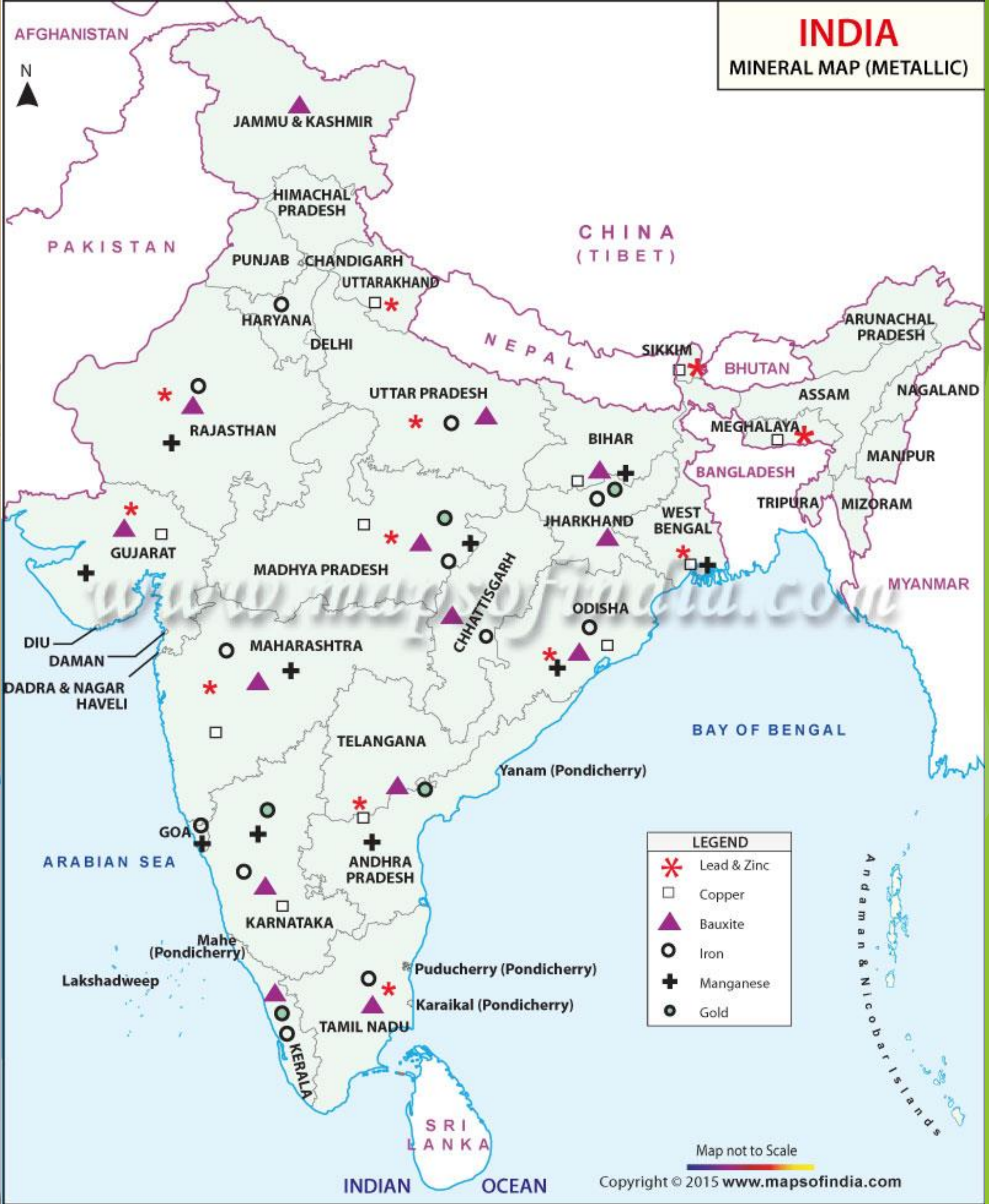
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SRI LANKA
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Andaman & Nicobar Islands
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Copper
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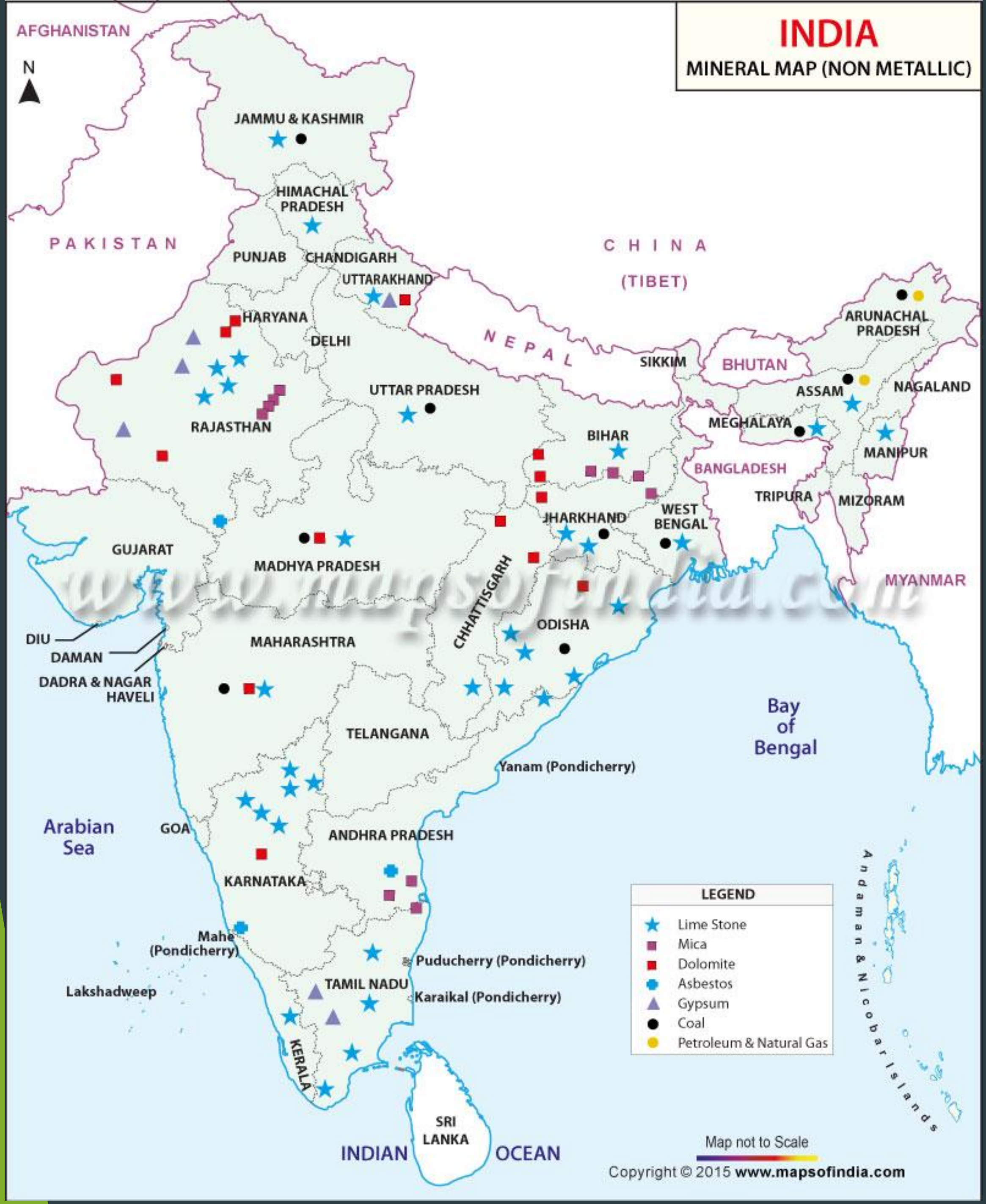
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MINERAL MAP (NON METALLIC)
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Petroleum & Natural Gas
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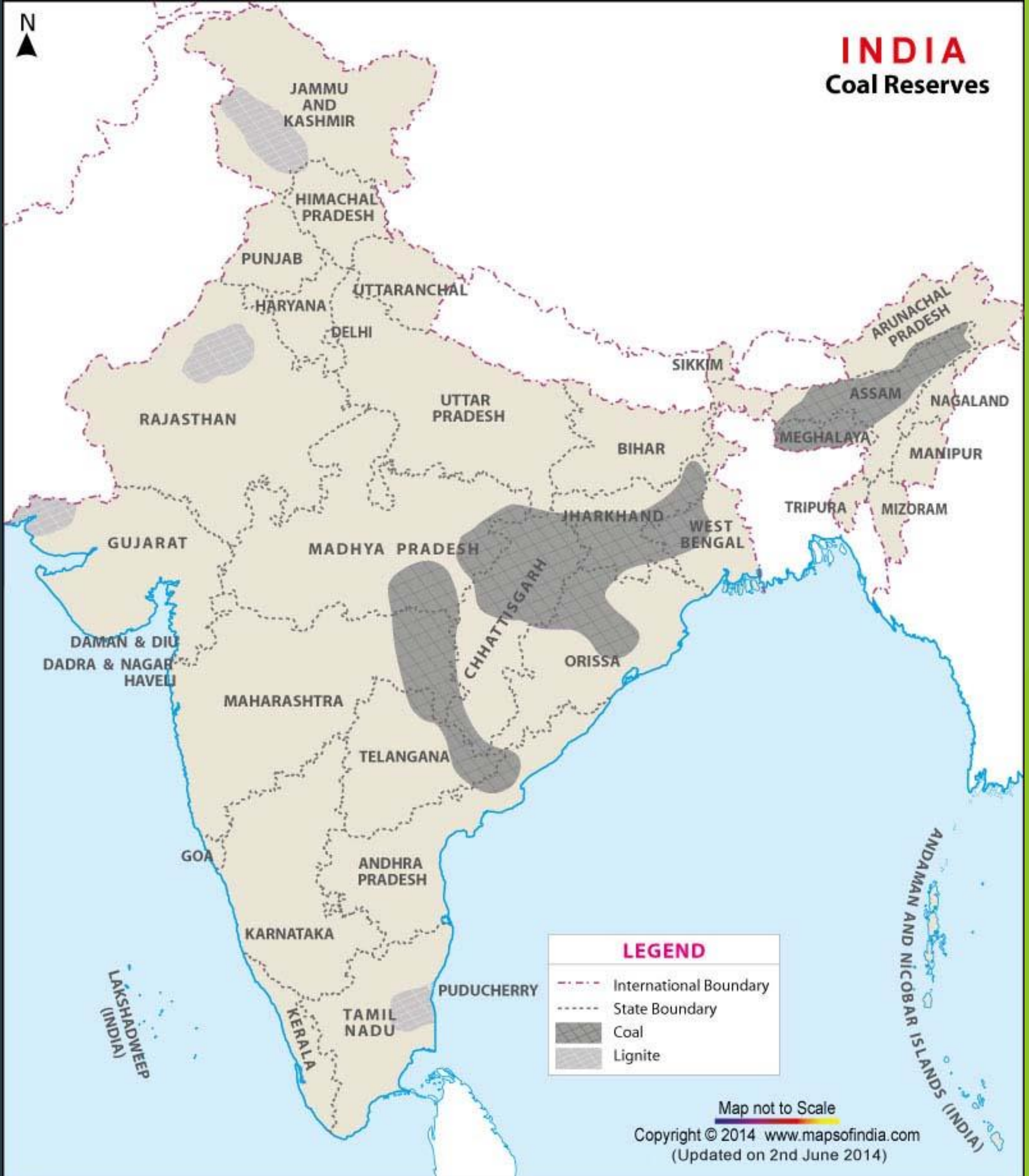
INDIA
Coal Reserves
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LAKSHADWEEP (INDIA)
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LEGEND
International Boundary
State Boundary
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Lignite
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Economic importance of the mineral industry

The **mining industry in India** is a major economic activity which contributes significantly to the economy of India.
The GDP contribution of the mining industry varies from 2.2% to 2.5% only but going by the GDP of the total industrial sector it contributes around 10% to 11%. Even mining done on small scale contributes 6% to the entire cost of mineral production. Indian mining industry provides job opportunities to around 700,000 individuals.

Risks in mining

Demand and Supply

- **Demand and supply analysis** is the study of how buyers and sellers interact to determine transaction prices and quantities.
- **Demand**, in economics, is the willingness and ability of consumers to purchase a given amount of a good or service at a given price.
- **Supply** is the willingness of sellers to offer a given quantity of a good or service for a given price.
- The demand and supply model is useful in explaining how price and quantity traded are determined and how external influences affect the values of those variables. Buyers' behavior is captured in the demand function and its graphical equivalent, the demand curve. This curve shows both the highest price buyers are willing to pay for each quantity, and the highest quantity buyers are willing and able to purchase at each price. Sellers' behavior is captured in the supply function and its graphical equivalent, the supply curve. This curve shows simultaneously the lowest price sellers are willing to accept for each quantity and the highest quantity sellers are willing to offer at each price.

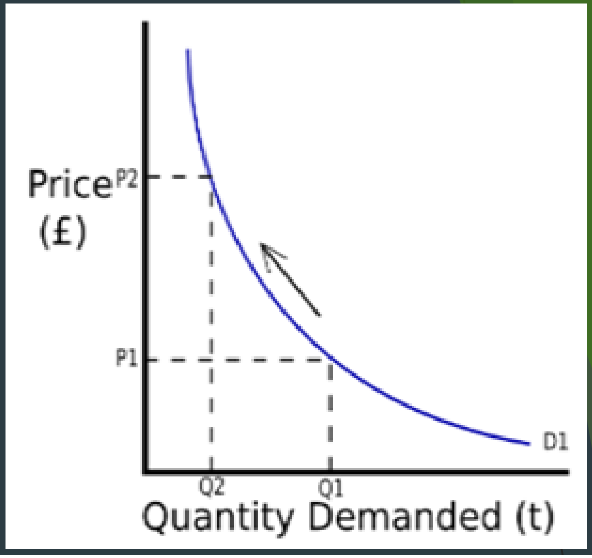
The Demand Function and the Demand Curve

- The quantity consumers are willing to buy clearly depends on a number of different factors called variables, Perhaps the most important of those variables is the item's own price. In general, economists believe that as the price of a good rises, buyers will choose to buy less of it, and as its price falls, they buy more.
- Although a good's own price is important in determining consumers' willingness to purchase it, other variables also have influence on that decision, such as consumers' incomes, their tastes and preferences, the prices of other goods that serve as substitutes or complements, and so on. Economists attempt to capture all of these influences in a relationship called the **demand function**. (In general, a function is a relationship that assigns a unique value to a dependent variable for any given set of values of a group of independent variables.) We represent such a demand function in Equation 1:

$$Q_x^d = f\left(P_x, I, P_y, \ldots\right) \qquad (1)$$

where Q^d_x represents the quantity demanded of some good X, P_x is the price per unit of good X, I is consumers' income, and P_y is the price of another good, Y

- In economics, a demand curve is a graph depicting the relationship between the price of a certain commodity and the quantity of that commodity that is demanded at that price. Demand curves can be used either for the price-quantity relationship for an individual consumer, or for all consumers in a particular market



Demand Function

Algebraically, the demand function can be represented as:

Qd = *f* (P, Ps, Pc, Y, A, A', N, CP, PE, TA, T/S*)*

- where Qd = quantity demanded of the good or service
- P = price of the good or service
- Ps = price of substitute goods or services
- Pc = price of complementary goods or services.
- Y = income of consumers.
- A = advertising expenditures (and other marketing expenditures)
- A' = competitors' advertising expenditures on the good or service.
- N = population (and other demographic factors).
- CP = consumer tastes and preferences for the good or service.
- PE= expected (future) changes in price.
- TA= adjustment time period.
- T/S = taxes or subsidies.

Demand and Supply as Equations

- Let's now look at demand and supply as equations. Here is a demand equation: ***Qd = 1,500 – 32P***
- To see what this equation says, we let price ***(P)*** *in the equation equal $10 and then solve* for quantity demanded ***Qd****. We get* ***Q_d = 1,180.***

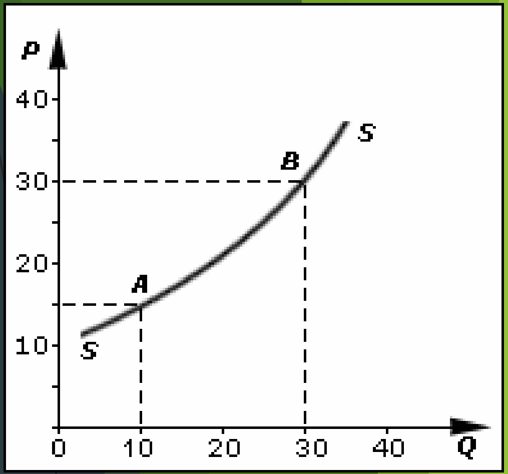
$$Q_d = 1{,}500 - 32(10) = 1{,}180$$

- So this equation says that if price is $10, it follows that quantity demanded is 1,180 units.
- We could find other quantities demanded by plugging in different dollar amounts for price (*P*).

$$Q_x^d = 8.4 - 0.4P_x + 0.06I - 0.01P_y \qquad (2)$$

The Supply Function and the Supply Curve

- The willingness and ability to sell a good or service is called supply. *In general, producers are willing to sell their product for a price as long as that price is at least as high as the cost to produce an additional unit of the product. It follows that the willingness to supply, called the* ***supply function, depends on the price at which the good can be sold as well as the cost of production for an additional unit of the good. The greater the difference between those two values, the greater is the willingness of producers to supply the good.***



For simplicity, we can assume that the only input in a production process is labor that must be purchased in the labor market. The price of an hour of labor is the wage rate, or *W. Hence, we can say that (for any given level of technology) the willingness to supply a good depends on the price of that good and the wage rate. This concept is captured in the following equation, which represents an individual seller's supply function:*

$$Q_x^s = f(P_x, W, \ldots)$$

SUPPLY FUNCTION

Supply function refers to the mathematical expression of the quantity supplied and the factors that determine the supply.

$$S_x = f(P_x\ P_1 \ldots. P_n\ C, T, G, O)$$

Where S_x refers to the quantity supplied of good X

P_x refers to the price of good X

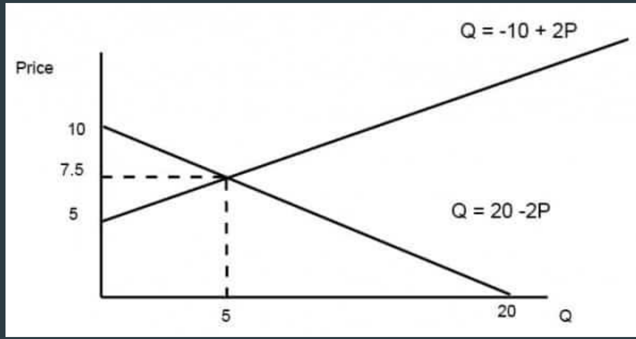
$P_1 \ldots. P_n$ refers to the price of related goods

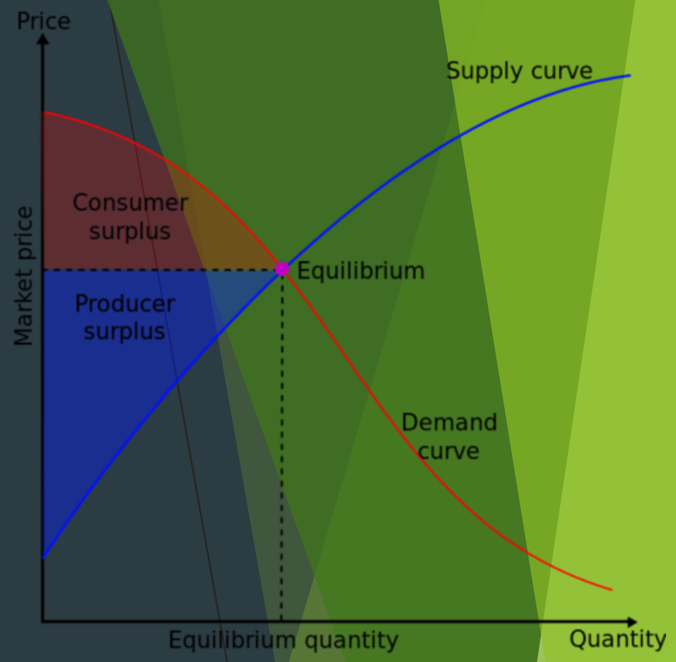
C refers to the cost of production

T refers to the state of technology.

G refers to the goal of the firm

O refers to other factors.





Elasticity of Demand

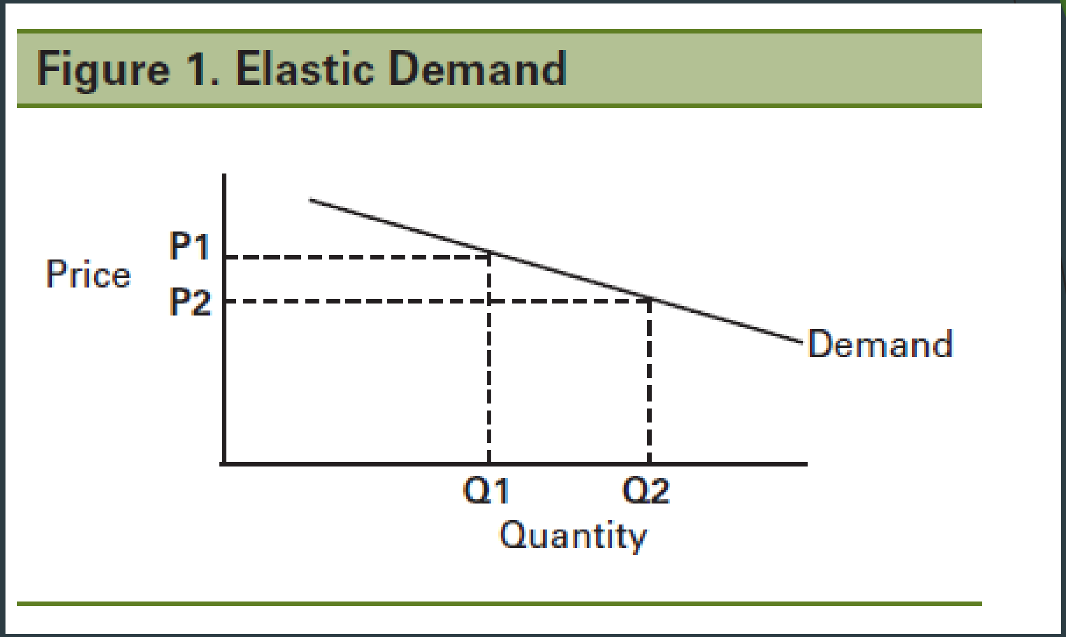
- **Elasticity of demand is an important variation on the concept of demand. Demand can be classified as elastic, inelastic or unitary.**
- An **elastic demand is one in which the change in quantity demanded due to a change in price is large. An inelastic demand is one in which the change in quantity demanded due to a change in price is small**
- The formula used here for computing elasticity of demand is:

$$\frac{(Q1 - Q2) / (Q1 + Q2)}{(P1 - P2) / (P1 + P2)}$$

$$E_S = \frac{\%\Delta Q}{\%\Delta P}$$

- If the formula creates an *absolute value greater than 1, the demand is elastic. In other words, quantity changes faster than price. If the value is less than 1, demand is inelastic. In other words, quantity changes slower than price. If the number is equal to 1, elasticity of demand is unitary. In other words, quantity changes at the same rate as price.*

Elastic Demand

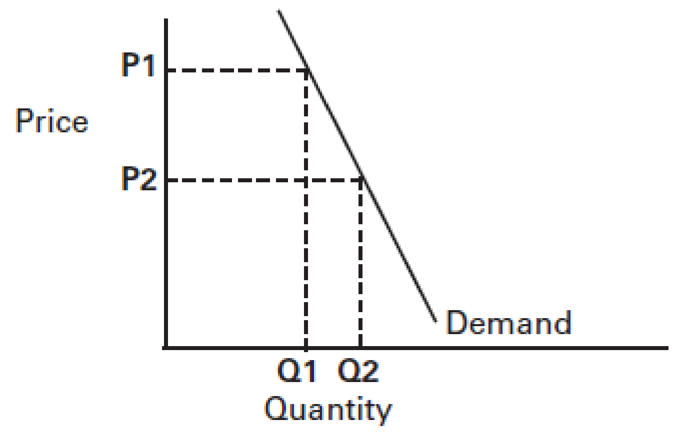


- When the price decreases from $10 per unit to $8 per unit, the quantity sold increases from 30 units to 50 units. The elasticity coefficient is 2.25.

Inelastic Demand

- Inelastic demand is shown in Figure 2. Note that a change in price results in only a *small change in quantity demanded. In other words, the quantity demanded is not very responsive to changes in price. Examples of this are necessities like food and fuel. Consumers will not reduce their food purchases if food prices rise, although there may be shifts in the types of food they purchase. Also, consumers will not greatly change their driving behavior if gasoline prices rise.*

Figure 2. Inelastic Demand

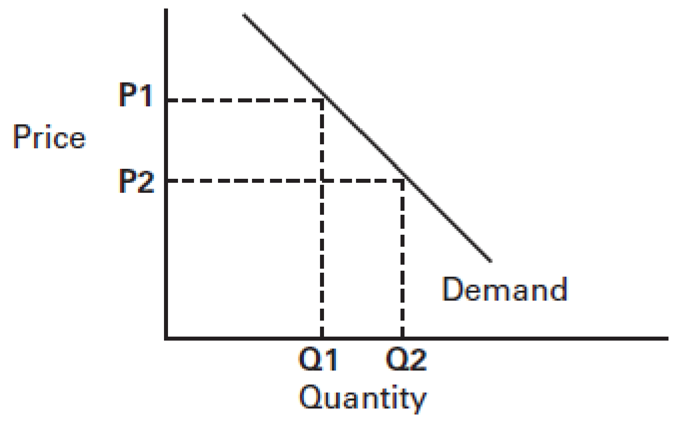


When the price decreases from $12 to $6 (50%), the quantity of demand increases from 40 to only 50 (25%). The elasticity coefficient is .33

Unitary Elasticity

- If the elasticity coefficient is equal to one, demand is unitarily elastic as shown in Figure 3. For example, a 10% quantity change divided by a 10% price change is one. This means that a 1% change in quantity occurs for every 1% change in price.

Figure 3. Unitary Elasticity



National Mineral Policy

- National Mineral Policy, 2019 has been approved by the Union Cabinet, on 28th February 2019.
- The aim of National Mineral Policy, 2019 is to have a more effective, meaningful and implementable policy that brings in further transparency, better regulation and enforcement, balanced social and economic growth as well as sustainable mining practices.
- The National Mineral Policy, 2019 includes provisions which will give boost to Mining Sector such as,
- Introduction of Right of First Refusal for RP/PL holders,
- Encouraging the Private Sector to take up exploration,
- Auctioning of virgin areas for composite RP-cum-PL-cum-ML on revenue share basis,
- Encouragement of merger and acquisition of mining entities
- Transfer of mining leases and creation of dedicated mineral corridors to boost Private Sector mining areas.

- proposes to grant status of industry to mining activity to boost financing of mining for Private Sector and for acquisitions of mineral assets in other countries by Private Sector
- Proposes to auction mineral blocks with pre-embedded clearances to give fillip to auction process.
- Propose to make efforts to harmonize taxes, levies & royalty with world benchmarks to help Private Sector
- The NMP-2019 will ensure more effective regulation. It will lead to sustainable Mining Sector development in future while addressing the issues of project affected persons especially those residing in tribal areas.

Thank you